Ch 7 problems

7.1 Straightforward

7.2

(7.12) $$\frac{t^*}{p} = \left[\frac{dm}{dp}\frac{p}{m}\right]^{-1}\left[\frac{dp^*/dt}{dp/dt}\right]$$

single foreign firm $\Rightarrow$ $x = m = d(p) = \alpha - \beta p \Rightarrow p = \frac{\alpha - x}{\beta}$

$$\pi^*(x) = x\cdot[p(x) - t] - C^*(x) \quad \text{noting } p^* = p(x) - t$$
$$= x\left[\frac{\alpha - x}{\beta} - t\right] - C^*(x)$$

$\Rightarrow$ FOC: $$\pi^{*\prime}(x) = \frac{\alpha}{\beta} - \frac{2x}{\beta} - t - C^{*\prime}(x) = 0$$

$$\Rightarrow x^* = \frac{\alpha}{2} - \frac{\beta}{2}\left[C^{*\prime}(x) + t\right]$$

$$\Rightarrow p = \frac{\alpha}{2\beta} + \frac{1}{2}\left[C^{*\prime}(x) + t\right]$$

$$\Rightarrow \frac{dp}{dt} = \frac{1}{2} \quad \text{and} \quad \frac{dp^*}{dt} = \frac{dp}{dt} - 1 = -\frac{1}{2}$$

So (7.12) $\Rightarrow$ $$\frac{t^*}{p} = -\left[\frac{dm}{dp}\frac{p}{m}\right]^{-1}$$

7.3

(7.22a) $\pi_x^* = p(z) + x\,p'(z) - [C^{*\prime}(x) + t] = 0$ where $z = x + y$

(7.22b) $\pi_y = p(z) + y\,p'(z) - C'(y) = 0$

Implicit differentiation of (7.22b) yields

$$\left.\frac{dy}{dx}\right|_r = -\frac{\pi_{yx}}{\pi_{yy}} = -\left[\frac{p'(z) + y\,p''(z)}{2p'(z) + y\,p''(z) - C''(y)}\right]$$

so $\frac{dy}{dx} < 0$ if $\underbrace{\pi_{yy} < 0}_{\text{true by SOC}}$ and $\underbrace{\pi_{yx} < 0}$

$\hookrightarrow$ requires $p'(z) + y\,p''(z) < 0$

Similarly we have, using (7.22a)

$$\left.\frac{dy}{dx}\right|_{r^*} = -\frac{\pi^*_{xy}}{\pi^*_{xx}} = -\left[\frac{p'(z) + x\,p''(z)}{2p'(z) + x\,p''(z) - C^{*\prime\prime}(x)}\right]$$

< 0 if $\underbrace{\pi^*_{xx} < 0}_{\text{by SOC}}$ and $\pi^*_{xy} < 0$

$\hookrightarrow$ requires $p'(z) + x\,p''(z) < 0$

7.4

(a) Using (7.22a), (7.22b)

$$\pi^*_{xx}\,dx + \pi^*_{xy}\,dy + \underbrace{\pi^*_{xt}}_{=-1}\,dt = 0 \qquad (1)$$

$$\pi_{yx}\,dx + \pi_{yy}\,dy = 0 \qquad (2)$$

$$(2) \Rightarrow \quad dx = -\frac{\pi_{yy}}{\pi_{yx}}\,dy \qquad (3)$$

$$\Rightarrow \pi^*_{xx} \cdot -\frac{\pi_{yy}}{\pi_{yx}}\,dy + \pi^*_{xy}\,dy = dt \qquad \text{using (1)}$$

$$\Rightarrow \frac{\pi^*_{xy}\pi_{yx} - \pi^*_{xx}\pi_{yy}}{\pi_{yx}}\,dy = dt$$

$$\Rightarrow \frac{dy}{dt} = \frac{-\pi_{yx}}{\underbrace{\pi^*_{xx}\pi_{yy} - \pi^*_{xy}\pi_{yx}}_{>0 \text{ by stability condition}}}$$

>0 if $\pi_{yx} < 0$ which requires $\pi_{yx} = p'(z) + y\,p''(z) < 0$

(b)

$$d\pi = \underbrace{\pi_y\,dy}_{=0 \text{ by FOC}} + \pi_x\,dx$$

$$\Rightarrow d\pi = \pi_x\,dx$$

$$\Rightarrow \frac{d\pi}{dt} = \underbrace{\pi_x}_{= y \cdot p'(z) < 0}\,\underbrace{\frac{dx}{dt}}_{<0} > 0$$

$$= \frac{-\underbrace{\pi_{yy}}_{<0 \text{ SOC}}\,\underbrace{\pi^*_{xt}}_{=-1}}{\underbrace{\pi^*_{xx}\pi_{yy} - \pi^*_{xy}\pi_{yx}}_{>0 \text{ by stability condition}}} \qquad \text{using (1),(2) above}$$

$$\underline{\underline{\text{OR}}} \quad = \underbrace{\frac{dx}{dy}}_{\substack{<0 \\ \text{by (3)} \\ \text{above}}}\,\underbrace{\frac{dy}{dt}}_{>0 \text{ by (a)}} < 0$$

7.5

(a) $y = f(v)$ h.o.d α $\Rightarrow f(\lambda v) = \lambda^{\alpha} f(v)$

So $\dfrac{\partial f}{\partial v} = \underbrace{\lambda f'(\lambda v) = \lambda^{\alpha} f'(v)}$

$\Rightarrow f'(\lambda v) = \lambda^{\alpha-1} f'(v)$

$\Rightarrow f'(v)$ h.o.d $\alpha - 1$

(b) $d_q(p,q,I)$ h.o.d -1 $\Rightarrow d_q(\lambda p, \lambda q, \lambda I) = \lambda^{-1} d_q(p,q,I)$

So $\eta(\lambda p, \lambda q, \lambda I) = -\lambda q \, d_q(\lambda p, \lambda q, \lambda I) / d(p,q,I)$

$= -\cancel{\lambda} q \cdot \lambda^{\cancel{-1}} d_q(p,q,I) / d(p,q,I)$

$= -q \, d_q(p,q,I) / d(p,q,I) = \eta$ h.o.d 0

(c) Given $d(p,q,I) = \phi(p,q) \cdot I$ then

$\eta(p,q,I) = \dfrac{-q \, \phi_q(p,q) \cdot \cancel{I}}{\phi(p,q) \cdot \cancel{I}} = \underline{\underline{\eta(p,q)}}$

Given η is h.o.d 0 then $\eta(\lambda p, \lambda q) = \eta(p,q)$ for <u>any</u> $\lambda > 0$.

So, letting $\lambda = 1/p$ we have $\underline{\underline{\eta(p,q) = \eta(1, q/p)}}$

7.6 Note:

a $\pi_p^* = \dfrac{d^*}{(1+\tau)} + \dfrac{p}{(1+\tau)} d_p^* - c^{*\prime} d_p^*$

$= d_p^* \left[\dfrac{p}{1+\tau} + \dfrac{d^*}{d_p^*(1+\tau)} - c^{*\prime} \right]$

$\boxed{\pi_p^* = d_p^* \left[\dfrac{p}{1+\tau} \left(1 - \dfrac{1}{\eta^*}\right) - c^{*\prime} \right]} \equiv d_p^* \cdot \underbrace{\text{FOC}}_{=0}$

$\Rightarrow \pi_{pp}^* = \underbrace{d_{pp}^* \cdot \text{FOC}}_{=0} + d_p^* \cdot \dfrac{1}{1+\tau} \left[1 - \dfrac{1}{\eta^*} + \dfrac{\eta^{*\prime}}{\eta^{*2}} \cdot \dfrac{p}{q} \right] \underline{\underline{> 0}}$

given $1 - \frac{1}{\eta^*} > 0$, $\eta^{*\prime} > 0$

$\pi_{pq}^* = \underbrace{d_{pq}^* \cdot \text{FOC}}_{=0} + d_p^* \cdot \dfrac{1}{1+\tau} \left[-\dfrac{\eta^{*\prime}}{\eta^{*2}} \cdot \left(\dfrac{p}{q}\right)^2 \right] \underline{\underline{< 0}}$

Implicit differentiation yields

$$\left.\frac{dp}{dq}\right|_{r^*} = -\frac{\overset{<0}{\Pi^*_{pq}}}{\underset{>0}{\Pi^*_{pp}}} \underline{\underline{>0}}$$

and $$\frac{dp}{dq}\cdot\frac{q}{p} = \frac{\frac{\eta^{*\prime}}{\eta^{*2}}\cdot\frac{p}{q}}{\underbrace{1-\frac{1}{\eta^*}}_{>0}+\frac{\eta^{*\prime}}{\eta^{*2}}\frac{p}{q}} \equiv B_1 \in \underline{\underline{(0,1)}}$$

$= \text{and} > 0$

(*) can show this is same as Feenstra's formula by dividing all terms by $1-\frac{1}{\eta^*} = \frac{\eta^*-1}{\eta}$

<u>Note</u>: $\Pi_q = d + q\,d_q - C'\cdot d_q$

$$= d_q\left[q + \frac{d}{d_q} - C'\right]$$

$$\boxed{\Pi_q = d_q\left[q\left(1-\frac{1}{\eta}\right) - C'\right]} \equiv d_q\cdot\underbrace{FOC}_{=0}$$

$$\Rightarrow \Pi_{qq} = \underbrace{d_{qq}\cdot FOC}_{=0} + d_q\cdot\left[1-\frac{1}{\eta}+\frac{\eta'}{\eta^2}\cdot\frac{q}{p}\right] \underline{\underline{>0}}$$

given $1-\frac{1}{\eta}>0$, $\eta'>0$

$$\Pi_{qp} = \underbrace{d_{qp}\cdot FOC}_{=0} + d_q\cdot\left[-\frac{\eta'}{\eta^2}\left(\frac{q}{p}\right)^2\right] \underline{\underline{<0}}$$

Implicit differentiation yields

$$\left.\frac{dp}{dq}\right|_r = -\frac{\overset{>0}{\Pi_{qq}}}{\underset{<0}{\Pi_{qp}}} \underline{\underline{>0}}$$

and $$\frac{dq}{dp}\cdot\frac{p}{q} = -\frac{\Pi_{qp}}{\Pi_{qq}}\cdot\frac{p}{q} = \frac{\frac{\eta'}{\eta^2}\cdot\frac{q}{p}}{\underbrace{1-\frac{1}{\eta}}_{>0}+\frac{\eta'}{\eta^2}\cdot\frac{q}{p}} \equiv B_2 \in \underline{\underline{(0,1)}}$$

$= \text{and} > 0$

b Implicit differentiation yields

$$\frac{dp}{d\tau} = -\frac{\pi^*_{p\tau}}{\pi^*_{pp}}$$

where using the FOC (1.28a)

$$\pi^*_{pp} = 1 - \frac{1}{\eta^*} + \frac{\eta^{*\prime}}{\eta^{*2}} \cdot \frac{p}{q}$$

$$\pi^*_{p\tau} = -C^{*\prime}$$

so

$$\frac{dp}{d\tau}\frac{(1+\tau)}{p} = \frac{\overbrace{C^{*\prime}\frac{(1+\tau)}{p}}^{=1-\frac{1}{\eta^*} \text{ by FOC}}}{1-\frac{1}{\eta^*}+\frac{\eta^{*\prime}}{\eta^{*2}}\frac{p}{q}} = \frac{1-\frac{1}{\eta^*}}{1-\frac{1}{\eta^*}+\frac{\eta^{*\prime}}{\eta^{*2}}\frac{p}{q}} \equiv B_3$$

$$\frac{dp}{d\tau}\frac{1+\tau}{p} = \frac{1}{1+\frac{p}{q}\cdot\frac{\eta^{*\prime}}{\eta^*(\eta^*-1)}}$$

c To solve for total changes in equilibrium values of p, q use the intersection of the reaction curves

$$q = r(p) = r(r^*(q,\tau)) \quad [\text{ie. } q^{eq} = r^{eq}(\tau)]$$

$$\rightarrow dq = r_p \cdot [r^*_q\, dq + r^*_\tau\, d\tau]$$

$$\Rightarrow \frac{dq}{q} = \underbrace{r_p \cdot \frac{p}{q}}_{=B_2} \cdot \left[\underbrace{r^*_q \cdot \frac{q}{p}}_{=B_1}\frac{dq}{q} + \underbrace{r^*_\tau \frac{(1+\tau)}{p}}_{=B_3}\frac{d\tau}{1+\tau}\right]$$

$$= B_2\left(B_1\frac{dq}{q} + B_3\frac{d\tau}{1+\tau}\right)$$

$$\Rightarrow \frac{dq}{d\tau}\frac{1+\tau}{q} = \frac{B_2B_3}{1-B_1B_2} \equiv B_4 \in (0,1)$$ upon substitution and rearranging

Note: this gives the equilibrium change in q in response to exogenous change in τ

$$p = r^*(q,\tau) \Rightarrow dp = r_q^* dq + r_\tau^* d\tau$$

$$\Rightarrow \frac{dp}{d\tau} = r_q^* \frac{dq}{d\tau} + r_\tau^*$$

so if we substitute in the equilibrium change in q then we get the equilibrium change in p

$$\Rightarrow \frac{dp}{d\tau}\frac{1+\tau}{p} = r_\tau^*\left(\frac{1+\tau}{p}\right) + r_q^* \frac{1+\tau}{p}\frac{dq}{d\tau}$$

$$= B_3 + r_q^* \frac{1+\tau}{p}\frac{dq}{d\tau}$$

$$= B_3 + r_q^* \frac{q}{p} \cdot \frac{dq}{d\tau}\frac{1+\tau}{q}$$

$$= B_3 + B_1 \cdot B_4$$

$$= B_3 + B_1\left(\frac{B_2 B_3}{1-B_1B_2}\right)$$

$$= B_3 + \frac{B_1B_2B_3}{1-B_1B_2}$$

$$= B_3\left[1 + \frac{B_1B_2}{1-B_1B_2}\right]$$

$$= B_3\left[\frac{1-\cancel{B_1B_2}+\cancel{B_1B_2}}{1-B_1B_2}\right]$$

$$= \frac{B_3}{1-B_1B_2} \in (0,1)$$ upon substitution and rearranging

7.1

(a) d is h.o.d 0 so

$$d(\lambda q, \lambda p, \lambda I) = d(q, p, I)$$

$$\Rightarrow \frac{\partial d(\lambda q, \lambda p, \lambda I)}{\partial \lambda} = d_q \cdot q + d_p \cdot p + d_I \cdot I = 0$$

$$\Rightarrow \underbrace{d_q \frac{q}{d}}_{\equiv -\eta} + d_p \frac{p}{d} + \underbrace{d_I \frac{I}{d}}_{\equiv 1} = 0$$

$$\Rightarrow d_p \frac{p}{d} = 1 - \eta$$

(b) Let's work with $\ln(y) = \ln(d(q, p, I))$

$$d\ln(y) = \frac{\partial \ln(y)}{\partial q} dq + \frac{\partial \ln(y)}{\partial p} dp + \frac{\partial \ln(y)}{\partial I} dI = 0$$

$$\frac{dy}{y} = \frac{1}{y}\frac{\partial d}{\partial q} dq + \frac{1}{y}\frac{\partial d}{\partial p} dp$$

$$\frac{dy}{y} = -\eta \frac{dq}{q} + (1-\eta)\frac{dp}{p}$$

$$\Rightarrow \frac{dy}{dp}\frac{p}{y} = -\eta \underbrace{\frac{dq}{dp}\frac{p}{q}}_{= B_q'} + (1-\eta)$$

$$= -\eta \left[\frac{\frac{\eta'}{\eta^2}\frac{q}{p}}{1 - \frac{1}{\eta} + \frac{\eta'}{\eta^2}\frac{q}{p}} \right] + 1 - \eta > 0$$

$$\Rightarrow \frac{\eta'}{\eta}\frac{q}{p} < (\eta - 1)\left[\frac{\eta - 1}{\eta} + \frac{\eta'}{\eta^2}\frac{q}{p}\right]$$

$$\Rightarrow \frac{\eta' q}{p}\left(1 - \frac{\eta - 1}{\eta}\right) < (\eta - 1)^2$$

$$\Rightarrow \frac{\eta'}{\eta}\frac{q}{p} < (\eta - 1)^2$$

7.8

(7.27a) $$\pi^* = \frac{p \cdot d^*(p,q,\tau)}{1+\tau} - c^*\left(d^*(p,q,\tau)\right)$$

$$\Rightarrow \frac{d\pi^*}{d\tau} = \underbrace{\pi^*_p}_{=0 \text{ FOC}} \frac{dp}{d\tau} + \pi^*_q \frac{dq}{d\tau} + \pi^*_\tau$$

$$= \left(\frac{p\, d^*_q}{1+\tau} - c^{*\prime} \cdot d^*_q\right) \frac{dq}{d\tau} - \frac{p\, d^*}{(1+\tau)^2}$$

$$\Rightarrow \frac{d\pi^*}{d\tau} \frac{1+\tau}{d^*} = \underbrace{\frac{q}{d^*} \cdot d^{q*}}_{\equiv \eta^* - 1} \underbrace{\left(\frac{p}{1+\tau} - c^{*\prime}\right)}_{= \frac{p}{\eta^*(1+\tau)} \text{ by FOC}} \frac{dq}{d\tau} \cdot \frac{(1+\tau)}{q} - \frac{p}{1+\tau}$$

$$= \frac{(\eta^* - 1)}{\eta^*} \cdot \frac{p}{1+\tau} \cdot \frac{dq}{d\tau} \frac{1+\tau}{q} - \frac{p}{1+\tau}$$

$$= \frac{p}{1+\tau}\left[\underbrace{\left(1 - \frac{1}{\eta^*}\right)}_{\in (0;1)} \cdot \underbrace{\frac{dq}{d\tau} \frac{1+\tau}{q}}_{< 1 \text{ by } 7.6(c)} - 1\right] \quad \underline{\underline{< 0}}$$

7.9

$$\max_{p_1} \quad \pi_1^*(p_1, q_1) + \delta\left[\theta \pi_2^*(\tau_2) + (1-\theta)\pi_2^*(0)\right]$$

$$\text{s.t } (1+\tau_2) = \frac{p^*}{p_1}$$

FOCs: $$\pi^*_p(p_1, q_1) + \delta\theta \underbrace{\pi_2^{*\prime}(\tau_2)}_{<0} \cdot \frac{p^*}{(p_1)^2} = 0 \quad \text{for foreign firm}$$

$$\pi_q(p_1, q_1) = 0 \quad \text{for home firm}$$

Given $\pi^*_{pp} < 0$ (soc) and $\pi_2^{*\prime} < 0$, the change in θ from $\theta = 0$ to $\theta > 0$ requires that $\underline{\underline{p_1 \text{ rise}}}$ to restore FOC = 0.